From Anecdotal Evidence to Quantitative Evaluation Methods: A Systematic Review on Evaluating Explainable AI

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The rising popularity of explainable artificial intelligence (XAI) to understand high-performing black boxes raised the question of how to evaluate explanations of machine learning (ML) models. While interpretability and explainability are often presented as a subjectively validated binary property, we consider it a multi-faceted concept. We identify 12 conceptual properties, such as Compactness and Correctness, that should be evaluated for comprehensively assessing the quality of an explanation. Our so-called Co-12 properties serve as categorization scheme for systematically reviewing the evaluation practices of more than 300 papers published in the last 7 years at major AI and ML conferences that introduce an XAI method. We find that 1 in 3 papers evaluate exclusively with anecdotal evidence, and 1 in 5 papers evaluate with users. This survey also contributes to the call for objective, quantifiable evaluation methods by presenting an extensive overview of quantitative XAI evaluation methods. Our systematic collection of evaluation methods provides researchers and practitioners with concrete tools to thoroughly validate, benchmark and compare new and existing XAI methods. The Co-12 categorization scheme and our identified evaluation methods open up opportunities to include quantitative metrics as optimization criteria during model training in order to optimize for accuracy and interpretability simultaneously.

CCS Concepts: • **Computing methodologies → Machine learning**; *Artificial intelligence*; • **General and reference → Evaluation**.

Additional Key Words and Phrases: explainable artificial intelligence, interpretable machine learning, evaluation, explainability, interpretability, quantitative evaluation methods, explainable AI, XAI

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SUPPLEMENTARY MATERIAL

Supplementary material presenting quantitative evaluation methods with user studies (Table 5), providing a more detailed discussion on the extent and nature of XAI research, and describing our paper collection and reviewing process in more detail.

7.1 Quantitative Evaluation with User Studies

Whereas the main paper focuses on automated quantitative evaluation methods, Table 5 summarizes the main quantitative evaluation methods we identified that were applied in user studies. Generally, we can distinguish between *subjective* evaluation and *objective* evaluation. The subjective methods usually evaluate Coherence by measuring how a user perceives an explanation. In contrast, the Forward Simulatability, Teaching Ability, Intruder Detection and Synthetic Artifact Rediscovery are objective evaluation methods. The Forward Simulatability method is comparable to the functionally-grounded Preservation Check: instead of evaluating whether the predictive model gives the right output given the explanation, the user acts as a surrogate model to evaluate whether the explanation is output-complete for a user. The Teaching Ability is a related approach but also evaluates whether the explanations are generalizable such that a user can, after being trained with explanations, make a correct prediction *without* having an explanation. We refer to other work (e.g. [55, 106, 142]) for a more detailed discussion on evaluation with human subjects.

7.2 Categorization and Analysis of XAI Methods

Figure 5 visualizes the categorization of papers per dimension. Generally, it can be seen that the top-3 in each dimension covers the majority of the papers. Especially the imbalance regarding the types of models to be explained is striking, since a large majority of the literature (75%) focuses on explaining neural networks, which could be due to their black box nature and state-of-the-art performance. The 'other' category follows with 17%, which usually involves models which are introduced in the paper and which are specifically designed to explain a certain prediction task. 15% of the papers are XAI methods which can, according to the authors, be applied to any predictive model. This model-agnostic category includes for example methods that explain a latent representation which can come from any model.

Our analysis also shows that there is a high diversity in explanation types. Interesting is that the top-3 explanation types are dominated by feature importance methods: 27% of the papers use standard feature importance scores, followed by heatmaps (2-dimensional feature importance) and localization (binary feature importance). The types of data that are given as input to predictive model f are also diverse, although images, text and tabular data are most used. Additionally, we found that image data make up the majority for heatmap explanations whereas textual input is mostly used for textual explanations. Feature importance seems to be the most generally applicable explanation type since it is used for all data types, and is also often used for 'Other' data types that do not fall into the predefined categories.

We also categorized the main task of the predictive models. These statistics might be highly influenced by our selection of publication venues and also the difficulty of a task can play a role. A great majority (63%) of the papers presents a model for classification. Besides the fact that classification is a broad concept, it can also be related with the popularity of outcome explanations: explaining a particular classification decision can be a good use case for outcome explanations. As shown in Figure 5, the type of problem that is addressed in 64% of the papers is the outcome explanation, meaning that an explanation aims to explain a single prediction. In contrast, the second most addressed type, model inspection (30% of the papers), gives global explanations about some property of predictive model f, such as global feature importance. This can imply that users or model developers are more interested in understanding

Table 5. Descriptions of quantitative evaluation methods with user studies, with references to papers that apply this method. Bold check mark indicates prominent Co-12 property.

Name and Description of Quantitative Metric, with References	Correctness	Output-completeness	Consistency	Continuity	Contrastivity	Covariate Complexity	Compactness	Composition	Confidence	Context	Coherence	Controllability
Forward Simulatability Given an explanation (and possibly the corresponding input sample), ask users to guess or identify the model's prediction (*human-output-completeness*). Additionally, the user's prediction speed can be measured, or the difference in simulation accuracy between whether or not explanations are shown. [7, 13, 18, 40, 42, 43, 98, 105, 131, 132, 143, 145, 154, 168, 176, 192, 212, 215–217, 220, 279, 282, 292]		✔		✓			✓	✓		✓		
Teaching Ability Train users with explanations to understand the model's reasoning, after which humans should predict the ground-truth for a new data instance *without* having an explanation. Additionally, the user's prediction speed can be measured. [88, 280]		✓		✓						✔		
Subjective Satisfaction Ask users to rate explanations on properties such as satisfaction, reasonableness, usefulness, fluency, relevance, sufficiency and trust. [3, 7, 24, 50, 51, 63, 81, 98, 124, 132, 140, 150, 168, 172, 192, 211, 217, 219, 233, 234, 259, 262, 276, 278, 285, 320, 325]						✓	✓	✓		✓	✔	
Subjective Comparison Show users explanations from different XAI methods (or explanations from humans) and evaluate which method is perceived as being better (in terms of e.g. perceived accuracy, usefulness or understandability). [18, 39, 49, 84, 119, 158, 177, 197, 242, 259, 276, 288]							✓	✓		✓	✔	
Perceived Homogeneity Ask users to evaluate the purity or disentanglement of explanations, by e.g. verifying that a dimension corresponds to a single interpretable factor. [240, 275, 313, 321]						✔					✓	
Intruder Detection Given an explanatory prototype or disentangled concept, show users a set of instances of which one is an intruder, and ask which instance does not correspond with the explanation. [84, 195, 251]						✔					✓	
Synthetic Artifact Rediscovery A controlled experiment where a property of the predictive model is changed, after which it is evaluated whether humans can reveal this property with the help of explanations. [218, 219, 245, 258]										✓	✔	

specific decisions than getting global insights, or that global XAI methods are more difficult to develop. The transparent box designs are by themselves already interpretable, such as a decision tree. Few papers present a model explanation, meaning that a second, interpretable model is learned to mimic the output of the black box. We hypothesize that the model explanation is not often addressed since there is no guarantee that the original black box and the interpretable surrogate model agree on their internal reasoning [227]. Hence, the resulting explanation could be incorrect with respect to the workings of the original predictive model. Interesting to add is that the outcome explanation is also the dominant problem for papers that do not introduce, but apply or evaluate an existing XAI method. Specifically, 82% of the papers excluded by the filter focus on outcome explanations. This might indicate that outcome explanations are easier to apply and compare among different XAI methods than global explanations.

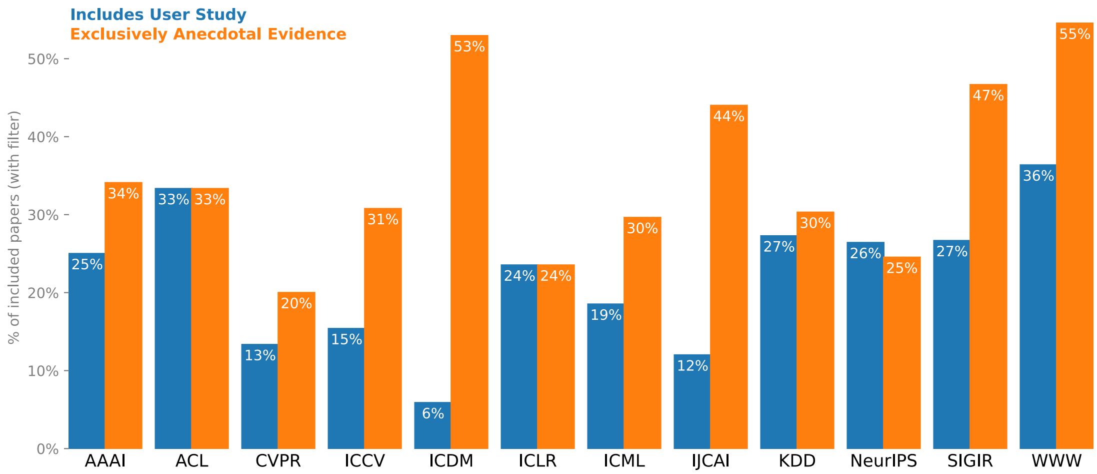

Fig. 8. Fraction of papers per venue that introduce an XAI method and evaluate with user studies (blue) or exclusively with anecdotal evidence (orange). The percentages are based on the total number of included papers per venue that introduce an XAI method.

Lastly, we see that the type of method used to explain is dominated by two types: post-hoc explanation methods that aim to explain an already trained model, and interpretability built into the predictive model. Built-in interpretability is a broad category ranging from models that are intrinsically interpretable to additions or restrictions to the predictive model architecture. The latter includes attention mechanisms, regularizers in the training process to improve interpretability, or a combined architecture that merges the prediction and explanation task. A minority produces explanations based on supervised explanation training, such as optimizing the XAI method to generate explanations that are similar to ground-truth explanations from a dataset. Interesting to note is that all supervised explanation training methods in our dataset produce outcome explanations.

7.2.1 XAI Evaluation Practice per Venue. Figure 8 provides more insight into the XAI evaluation practices per venue. It shows that generally the more application-oriented conferences evaluate only with anecdotal evidence in roughly half of the cases. In contrast, the more theoretical venues have percentages around 20%-30%. We could not see a clear trend regarding the usage of user studies, although the relative differences between conferences is striking.

7.3 Methodology

This section describes in more detail our paper collection process, inclusion and exclusion criteria, and reviewing process.

7.3.1 Identification of Paper Candidates. We collected papers in a structured manner to provide both quantitative and qualitative insights about the XAI domain based on a large corpus of scientific work on XAI evaluation methods. Since the literature on XAI is highly diverse and distributed across different (sub)disciplines, we selected literature by filtering on publication venue, date and title.

Publication Venue. To obtain a representative and sufficiently large, yet feasible selection of papers, we considered literature from all areas of AI, ranging from Computer Vision and Information Retrieval to Natural Language Processing and Data Mining. Specifically, we considered literature from the following twelve prominent[3] conferences: AAAI, IJCAI, NeurIPS (formerly NIPS), ICML, ICLR, CVPR, ICCV, ACL, WWW, ICDM, SIGKDD (also called KDD), SIGIR. This selection criterion also implies that all work included in the set is original, peer-reviewed and written in English. We are aware of the fact that we exclude relevant papers published at other venues, but do believe that our selection of venues is sufficiently representative to enable extrapolation of our results and conclusions to XAI literature as a whole.

Publication Year. We scoped our selection to work published from 2014 to 2020. This criterion is motivated by the fact that XAI has gained renewed interest since the emergence of deep learning, and the fact that annual international conference series dedicated exclusively to explainability or interpretability were organized from 2014 onwards [1].

Keywords in Title. We conducted a keyword search in publication titles with the following search query: `explain* OR explanat* OR interpret*` to capture terms including *explainable*, *explaining*, *explanation*, *interpretable* and *interpretability*. We are aware of the fact that this query excludes papers with related terms (such as *intelligibility* and *transparency*), and papers that specify a specific explanation method (such as *feature attribution*). However, to reflect time and resource constraints, we aimed for high precision instead of high recall. For a similar reason, we did not consider keywords in abstracts or full-texts, since we found that searching for general terms as *explain* leads to a surge in irrelevant results.

Final Search Query. We used the search engine of computer science bibliography DBLP[4] to collect the initial selection of papers. Combining the aforementioned criteria results in the following query to DBLP:

```
explain | explanat | interpret year:2020: | year:2019: | year:2018: | year:2017: | year:2016: | year:2015: | year:2014: venue:ICDM: | venue:KDD: | venue:NIPS: | venue:NeurIPS: | venue:CVPR: | venue:ICCV: | venue:AAAI: | venue:IJCAI: | venue:SIGIR: | venue:ACL: | venue:WWW: | venue:ICLR: | venue:ICML:
```

This search, conducted on 4th of May 2021, resulted in 606 papers.

7.3.2 Inclusion and Exclusion. Before screening the main content of each paper according to inclusion criteria, we applied an exclusion criterion since we found that the search result from the query contained more than the main conference papers.

Exclusion. We manually excluded companion papers, which include extended abstracts and papers from workshops, doctoral consortium and early career tracks, invited talks, senior member presentations, demonstrations, companion proceedings, challenges and tutorials. Applying this exclusion criterion to the initial query result resulted in 494 papers.

Inclusion. The resulting 494 papers are screened according to an inclusion criterion, in order to only include relevant papers in our analysis. With our inclusion criterion, we focus on papers in the explainable AI domain and therefore exclude papers that use the terms "explain" or "interpret" in other contexts.

Inclusion Criterion
Original work introducing, applying and/or evaluating one or more methods for explaining a machine learning model.

With "introducing", we mean that the work presents a new method for explaining a machine learning model. The term "machine learning" implies learning from data. Since we require that this machine learning model should be explained (see the main paper for our definition of 'explanation'), we do not include papers that only explain the *data* rather than explaining how a predictive model does something.

[3] As indicated by their A* ranking according to the CORE 2021 rankings portal (http://portal.core.edu.au/conf-ranks/).
[4] https://dblp.org/

Applying the inclusion criteria to the set of 494 papers, led to 361 papers being included. Subsequently, we can apply a **filter** that only selects the papers that *introduce* an XAI method, resulting in 312 papers. We apply this filter to analyze how introduced XAI methods are evaluated when they are first presented. For collecting all evaluation methods, we review all 361 included papers since 49 papers do not introduce a new XAI method, but could contain relevant evaluation metrics to compare and evaluate existing XAI methods. We do not want such papers to skew our quantitative results on XAI methods (Figure 5), but include them in our evaluation overview for completeness.

7.3.3 Inclusion and Reviewing Process. Screening papers for inclusion and reviewing them according to a review protocol was a collaborative task. All authors have a background in machine learning and have explainable AI as research interest. Each author reviewed papers individually but there was frequent communication within the team to align and verify inclusion and categorization decisions. For 81 papers, the inclusion criteria were checked by two reviewers in order to measure the inter-rater agreement for quality assurance. Besides a random sample of papers that were reviewed twice, the majority in this set were papers where the initial reviewer indicated that they were not confident about the decision, after which another reviewer checked the paper. We therefore emphasize that the following agreement metrics can be biased towards lower scores due to the perceived difficulty of the papers in the specific subset. The two reviewers were in agreement on the inclusion decision for 68 out of 81 papers, resulting in a Cohen's kappa $\kappa = 0.625$ and a Matthews correlation coefficient $MCC = 0.647$ (the latter is said to be better suited for binary and imbalanced data [54]). These results indicate substantial agreement [148]. In case of disagreement or low confidence by both reviewers, discussion took place to come to a final inclusion decision. In addition to disagreement, also more informal discussion took place whenever a reviewer was in doubt on aspects of the review protocol. After reviewing, the first author did an extra check regarding the categorization of evaluation methods of all included papers.

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